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T2D-MAS: From Software Ticket to Deployment

What the prototype does, how its agents work, and what the experiment shows.

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The project targets four delivery bottlenecks

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Original problem statement from the project report and idea document

- **Manual triage slows resolution**
  Severity, category and team routing depend on people reading each ticket.
- **Approvals become queues**
  Human review is essential for risky actions, but reviewers need useful context.
- **Duplicate reports create repeated work**
  Similar incidents can be phrased differently, so keyword matching misses them.
- **Quality varies between tickets**
  Reproduction steps, acceptance criteria and release notes can be incomplete.

Goal

Coordinate specialized software agents from user feedback through deployment, while keeping people in control of high-risk decisions.

Every stage now runs; real systems are simulated

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Use this distinction when interpreting every capability and result

ORIGINAL IDEA

A coordinated delivery team

Specialist agents handle triage, investigation, planning, implementation, QA, deployment and monitoring. ML, deep learning, LLMs and feedback loops support the full lifecycle.

WORKING REPOSITORY

Nine agents, end to end

Real: feedback masking, duplicate search, triage, the Contract Net auction, mobile trips to six data nodes, six gates and the audit log. Simulated with seeds: project board, repository, CI, deployment and telemetry.

OpenRouter (free models only) is optional for triage and reviewer notes. The benchmark stays deterministic; model proposals cannot approve gated actions.

An agent is a role with a controlled action loop

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The design assigns work to specialists; the code defines a shared five-part contract

1 **Perceive input** › 2 **Remember small state** › 3 **Reason proposal** › 4 **Act behind gate** › 5 **Report auditable result**

Roles in the delivery team

Scouts read feedback, visit the data nodes and watch releases. Brokers check duplicates, judge impact and run the squad auction. Workers investigate, plan, draft, test and release. Security runs six gates; a person approves, asks for changes or rejects.

A role describes responsibility. Mobility describes where an agent runs. A0–A3 describe benchmark architectures.

A0–A3 change the coordination architecture

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All four arms use the same seeded synthetic tickets and five seeds

ARM	A0	A1	A2	A3
Architecture	Rules, one process	One agent	Multiple static agents	Multi-agent + mobile agent
Coordination	No agent messages	One agent loop	Agent messages between roles	Messages + signed bundle to node
Work location	Central process	Central process	Agents run centrally	Serialized mobile bundle at investigation
Measured mean bytes / ticket	0	1,778	3,556	4,161
Mean messages / ticket	0	6	12	13

The dashboard's Run a ticket path and its node trips are separate from the A0–A3 benchmark.

Run a ticket: nine agents from feedback to release

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Real decisions and gates; the repository, CI, deploys and telemetry are seeded simulators

1 Feedback capture

Mask PII · language · version

2 Triage

Duplicates · severity · gate

3 Assignment

Contract Net: three squads bid

4 Investigation

6 node trips · suspect ranking

5 Planning

Subtasks · Given/When/Then

6 Implementation

Draft PR · secret scan

7 QA

About 100 of 3,400 tests · NAPFD

8 Deployment

Risk score → 5% canary → release

9 Monitoring

SLO check → follow-up ticket

Merge and release always wait for a person; four more gates stop only when their rule fires.

Example: one flash-sale complaint, end to end

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FB-001, a casual Indonesian crash report, waits for a person at triage, merge and release

1 · Intake

Indonesian report, PII masked

2 · Triage

Critical · 0.84 confidence

3 · Assign

Payments squad wins (U = 0.87)

4 · Investigate

6 nodes · payment-db first

5 · Plan

Subtasks + Given/When/Then

6 · Implement

Draft PR passes secret scan

7 · QA

About 100 tests; merge review

8 · Deploy

Canary, then release sign-off

9 · Monitor

Healthy vs SLO; ticket closed

Critical risk (1.0) is above R_max = 0.60, so triage waits for a person; merge and release always do. A repost of the complaint (FB-002) scores 0.956 and is auto-linked.

AI integration is hybrid, with a bounded optional LLM

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The interactive path, benchmark and separate ML modules use different methods

Severity triage	English and Indonesian rules by default. An optional free OpenRouter model proposes severity; its confidence is capped at 0.69, so a person still reviews.
Review feedback	Optional OpenRouter structures a note into changes, constraints and evidence. It can ask for clarification; reviewer decisions stay human.
Duplicate search	Live TF-IDF cosine over character n-grams. It reproduces the idea's A/B ≈ 0.994 (linked) and A/C ≈ 0.119 (new ticket).
Separate models	Benchmark: TF-IDF + calibrated LinearSVC. Offline: MiniLM + FAISS, DistilBERT, trained GraphSAGE. Live: the GraphSAGE-mean reference ranking.
Separate policies	Contract Net assigns squads. PPO/DQN stay offline in CI/canary simulations. No code-writing model; CI and deploys are seeded simulators.

Benchmark: no hosted LLM. Dashboard runs: an optional free OpenRouter model when a key is set.

A2 vs A3: static and mobile execution

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Mobility is about moving a signed agent to the data, not adding another role

A2 · STATIC

Agents execute centrally. The node sends data or evidence back to the coordinator.

A3 · MOBILE

A signed bundle runs at a node. The node keeps raw data local and returns allowlisted aggregates.

Example exchange

Static agent → sends query

Mobile bundle → node

Aggregate → coordinator

In the dashboard, Scout-Log makes these trips during investigation: Ed25519-signed bundles, an egress allowlist and a 42 MB break-even per node. A3 adds bundle bytes in the benchmark.

The benchmark result set is synthetic

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The repository can load public bug data, but the reported A0–A3 experiment used generated tickets

A0–A3 benchmark

500 tickets per arm: 100 generated tickets × 5 seeds (42–46). Version: synthetic-eval-v1. The same generated cases feed all four arms.

Ticket fields

ID · title · body · component · severity · creation time · duplicate link · fault label

Optional source datasets

Loaders support GitBugs and Eclipse/Mozilla issue exports. Preprocessing masks common PII patterns, collapses repeated stack-trace lines and creates chronological 70/10/20 train, validation and test splits.

Interpretation

Benchmark split: first 70% train, final 30% evaluate. General data loader split is 70/10/20. DORA-style outcomes are simulated; token use is estimated.

A1–A3 improve synthetic retrieval and fault ordering

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Run cited in the report · 500 generated tickets per arm · host timings are only workload proxies

Metric	A0	A1	A2	A3
Mean wall time / ticket	0.102 ms	0.155 ms	0.184 ms	0.198 ms
Network bytes / ticket	0	1,778	3,556	4,161
Duplicate Recall@5	0.429	1.000	1.000	1.000
Duplicate MAP	0.242	0.680	0.687	0.685
Severity macro-F1	1.000	1.000	1.000	1.000
Fault ordering NAPFD	0.490	0.750	0.750	0.750
Simulated rollback rate	5.6%	5.4%	5.8%	7.4%

A3 adds the most bytes and has the highest rollback rate. All arms reach F1 = 1.0 on distinctive synthetic labels.

Treat these as classroom results, not field evidence

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Seeded results support discussion of trade-offs; they do not validate business impact

- **Wall time differed in this run**
  Friedman omnibus p = 0.0136 across five seeds. Values remain tiny CPU-proxy timings, not service latency.
- **Policy gates held in this run**
  20/20 synthetic injections blocked; zero unapproved full releases. A3 had 11 approved simulated releases.
- **Network bytes also differed**
  Friedman omnibus p = 0.0018. CPU (p = 0.926) and rollback (p = 0.604) were not significant.
- **No field productivity claim**
  The harness runs no LLM API, Redis, real CI or production deployment. DORA-style outcomes are simulated.

What the result actually says

Coordination and mobility increase serialized work in this setup. Whether local execution is worth that cost depends on data locality and privacy requirements, which this benchmark does not quantify.

Conclusion: a useful prototype, not autonomous delivery

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The implemented boundary is clear, and it gives the next evaluation a practical target

What works today

Nine agents take natural-language feedback to a monitored release: live duplicate search, a squad auction whose winner investigates, signed node trips, six gates and a monitoring loop.

What the experiment shows

Agent arms improve duplicate retrieval and fault ordering in generated data. A3 adds transfer cost; the interactive dashboard is a different path, and real data locality is not measured.

Conclusion

T2D-MAS shows inspectable agent collaboration with people in control. Next, replace the simulators with a real repository, CI, release pipeline and telemetry, move the offline models into the live path, and evaluate on public data.

Keep the boundary clear: models propose, humans decide, irreversible actions remain gated.

Sources and scope notes

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Repository and supplied project documents checked for this deck

- Project 1 - Idea.pdf — original problem, illustrative cosine example, agent design and AI capability roadmap.
- Laporan_Proyek1_T2D-MAS_Final.pdf — expanded design, agent taxonomy, methods, intended workflow and the cited results (§9.4–9.5).
- README.md; docs/AUDIT.md; docs/process_and_experiments.md; agents/AGENT_CATALOG.md — architecture, pipeline and what is real, simulated or not built.
- agents/pipeline.py; scouts.py, brokers.py, workers.py; policy.py; severity_triage.py; deliverables/T2D-MAS-Live-Demo-Guide.md — the nine agents, their gates, optional OpenRouter and the demo walkthrough.
- eval/runner.py; eval/config.json; results/20260930T020846305626/summary.csv, statistical_tests.csv and security_injection_summary.csv — the benchmark outputs the report cites.
- data/README.md; triage/README.md; rca/README.md; envs/README.md — dataset loaders and separate model/simulation modules.

A fresh run matches the cited CSVs on every non-timing value (eval.compare_runs). Timing, simulation and estimate labels remain explicit.

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Thank you.
Code, data and results:
github.com/zidanarsyad/t2d-mas

